

GOTTA Prototype: Known-Asteroid Extraction and Statistical Performance Evaluation

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Abstract The GOTTA Prototype is a wide-field, high-cadence pathfinder for the full Global Open Transient Telescope Array. Its repeated survey imaging naturally records large numbers of Solar System small bodies, providing an opportunity to validate moving-object processing before the full array is deployed. We present a known-object association workflow for recovering cataloged asteroids from GOTTA Prototype photometric catalogs. For each exposure, the workflow determines the CCD footprint and mid-exposure epoch, predicts known-object ephemerides, filters candidates within the detector footprint, associates predicted positions with calibrated catalog detections, rejects likely stationary-source contaminants using a Gaia cross-check, and augments the recovered detections with orbital and observing-geometry information from external small-body databases. The resulting sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including 77 near-Earth objects and 19 potentially hazardous asteroids. The sample is dominated by main-belt asteroids. The observed-minus-predicted positional residual has a median separation of 0.220 arcsec and a two-dimensional RMS of 0.355 arcsec. We characterize the recovered sample in sky position, orbital class, aperture photometry, astrometric residual, angular rate, observing geometry, and repeated temporal sampling. As a pilot light-curve

application, we combine selected GOTTA detections with auxiliary 60 cm Schmidt observations and identify 12 reliable rotation-period measurements, together with 12 additional tentative candidates requiring follow-up. These results demonstrate that the GOTTA Prototype has established a working known-object recovery layer and provide a practical baseline for future moving-object association, auxiliary light-curve follow-up, NEO follow-up support, and Solar System studies with the full GOTTA array.

Key words: surveys — minor planets, asteroids: general — techniques: photometric — techniques: astrometric — methods: data analysis

1 INTRODUCTION

Near-Earth objects (NEOs) and potentially hazardous asteroids (PHAs) have long motivated systematic searches for small bodies in the Solar System. The Spaceguard Survey emphasized that the first step toward mitigating impact hazards is a comprehensive search for Earth-crossing asteroids and comets, followed by reliable orbit determination and long-term tracking (Morrison 1992). During the past three decades, wide-field CCD surveys have transformed this task from targeted searches into survey-scale operations. Programs such as Spacewatch, LINEAR, the Catalina Sky Survey, Pan-STARRS, and ATLAS have greatly increased the inventory of known minor planets and have established the basic observing and processing framework for modern moving-object surveys (e.g. Stokes et al. 2000; Larsen 2007; Larson et al. 2003; Denneau et al. 2013; Tonry et al. 2018).

Moving-object science in time-domain surveys depends not only on single-image depth, but also on cadence, sky coverage, astrometric precision, and the ability to associate detections across time. A discovery-oriented pipeline generally starts from transient or non-stationary detections, forms same-night tracklets, and links tracklets obtained on different nights to determine preliminary orbits (Kubica et al. 2007; Denneau et al. 2013; Vereš & Chesley 2017). The same data stream also supports attribution and precovery of known objects, which are essential for orbit refinement, survey validation, and physical characterization. This distinction is particularly important for prototype surveys: even before a full discovery pipeline is finalized, the recovery of cataloged asteroids can be used to evaluate astrometric performance, photometric behavior, source extraction quality, and temporal sampling under real observing conditions.

Several recent and forthcoming surveys illustrate the range of strategies relevant to Solar System time-domain work. ATLAS was designed as a high-cadence all-sky system optimized for the discovery of dangerous near-Earth asteroids and other rapidly varying sources (Tonry et al. 2018). ZTF combines a 47 deg² camera with near-real-time reduction and alert distribution for transients, variables, and moving objects (Bellm et al. 2019; Patterson et al. 2019). The Rubin Observatory Legacy Survey of Space and Time will provide much deeper multi-epoch imaging and is expected to substantially expand the number of Solar System detections (Ivezić et al. 2019; Jones et al. 2018;

[Schwamb et al. 2023](#)). In parallel, the NEO Surveyor mission will search in the infrared from space and is optimized for discovering and characterizing NEOs, especially objects difficult to observe from the ground ([Mainzer et al. 2023](#)). These facilities show that cadence, depth, wavelength, and sky coverage are complementary rather than interchangeable survey properties.

High-cadence survey data are also valuable for asteroid physical characterization. Time-series photometry constrains rotation periods and amplitudes, and multi-geometry light curves can be used to infer spin states and shapes through light-curve inversion ([Kaasalainen & Torppa 2001](#); [Kaasalainen et al. 2001](#); [Warner et al. 2009](#)). Sparse survey photometry further enlarges the statistical sample of asteroid light curves when sufficient temporal and phase-angle coverage is available. Accurate photometry and observing geometry are therefore necessary not only for detection and orbit association, but also for studying asteroid rotation, shape, surface properties, and activity. Phase-angle effects must also be treated explicitly when survey photometry is interpreted physically, for example through the H, G_1, G_2 phase-function formalism ([Muinonen et al. 2010](#)).

The SiTian/GOTTA program is designed to provide high-cadence, wide-field optical monitoring with a distributed array of telescopes ([Liu et al. 2021](#)). Its pathfinder, the Mini-SiTian Array at Xinglong Observatory, has already demonstrated the scientific value of high-cadence, multi-band observations for time-domain astronomy and has provided operational experience for the full SiTian/GOTTA system ([Huang et al. 2025](#); [Han et al. 2025](#)). The Mini-SiTian Array consists of three 30 cm telescopes, has operated at Xinglong since 2022, and provides large-field photometric monitoring that informs both observation strategy and data processing for the full SiTian/GOTTA system. These pathfinder data and procedures are directly relevant to the development of Solar System analysis tools for future high-cadence survey operations.

For asteroid science specifically, [Liu et al. \(2025\)](#) developed an asteroid photometry and period-analysis pipeline using Mini-SiTian data from the f02 region. They derived rotation periods for 22 asteroids, including 14 objects without periods listed in ALCDEF, and found agreement with existing results for several objects with sufficiently dense photometric sampling. This work demonstrates the feasibility of asteroid photometry with SiTian pathfinder observations and provides an important motivation for building the corresponding known-object association and validation layer for the GOTTA Prototype.

The GOTTA Prototype extends this pathfinder context to a larger-aperture wide-field system. Mini-SiTian has shown that small pathfinder facilities can deliver useful asteroid light curves, while the GOTTA Prototype provides a testbed for survey-scale source catalogs, exposure geometry, and small-body database augmentation. The present work does not attempt to report unknown asteroid discoveries, measure formal completeness, or operate a full moving-object linking system. Instead, it focuses on known-object recovery, a necessary precursor to tracklet construction, moving-object alert filtering, orbit refinement, and light-curve preparation.

In this paper we present a known-asteroid extraction pipeline for GOTTA Prototype photometric catalogs. We use cataloged minor planets as a controlled sample to validate the end-to-end association between survey catalogs and known-object ephemerides. This approach allows us to evalu-

ate astrometric residuals, photometric measurement behavior, angular-rate coverage, orbital-class distribution, and repeated-observation statistics. The recovered sample contains 56,230 matched detections of 16,053 unique known asteroids, including a small but useful subset of NEOs and PHAs. We also include a pilot light-curve analysis using auxiliary 60 cm Schmidt observations to demonstrate how the recovered known-object catalog can feed into an asteroid period-search workflow.

The paper is organized as follows. Section 2 describes the GOTTA Prototype data and the recovered known-asteroid sample. Section 3 presents the ephemeris-based extraction and augmentation method. Section 4 analyzes the recovered sample and evaluates the association performance. Section 5 describes the light-curve analysis framework and the role of auxiliary observations, and Section 6 discusses the implications for future GOTTA moving-object processing and summarizes the conclusions.

2 OBSERVATIONS AND CATALOG DATA

2.1 GOTTA Prototype Survey Data

The observations analyzed in this work were obtained with the GOTTA Prototype during the original four-CMOS test campaign from 2025 January to June. The prototype is a 1 m-class wide-field Ring Schmidt system located at Xinglong Observatory and was operated during this stage with a single SDSS-like g filter. The optical design, detector layout, and observing configuration are described in detail by Zhengyang Li et al. (in preparation); here we only summarize the aspects required for known-asteroid recovery.

The present analysis uses calibrated source catalogs produced by the GOTTA Prototype basic data-reduction pipeline. That pipeline performs calibration-frame construction, image preprocessing, astrometric calibration, photometric calibration, and source extraction, as described by Niu Li et al. (in preparation). The astrometric solution is tied to Gaia DR3 (Gaia Collaboration et al. 2023), and the photometric catalog provides source positions, aperture photometry, Kron-like photometry, photometric uncertainties, and quality flags for each exposure. Related CMOS wide-field reduction strategies have also been developed for Mini-SiTian (Xiao et al. 2025). For known-asteroid recovery, the most important quantities are the WCS, the exposure time and observing epoch, the measured source coordinates, and the aperture-photometry measurements.

The companion reduction analysis shows that the calibrated catalogs are sufficiently accurate for sub-arcsecond source association. Stellar astrometric residuals are much smaller than the matching scale used here, while the photometric catalog includes aperture-corrected measurements in a sequence of circular apertures. Asteroid detections are generally fainter than the stars used for calibration and may be affected by motion during the exposure. Their observed-minus-predicted residuals therefore include additional contributions from centroiding uncertainty, ephemeris uncertainty, timing uncertainty, and possible trailing.

2.2 Recovered Known-Asteroid Sample

The recovered sample consists of accepted associations between predicted positions of cataloged asteroids and calibrated GOTTA source detections. Each association records the measured source position and photometry, the predicted ephemeris, the observing epoch, and auxiliary orbital and observing-geometry quantities. The final sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights. It is dominated by main-belt asteroids, with smaller contributions from Mars-crossing asteroids, Jupiter Trojans, NEOs, and PHAs. A subset of objects with repeated detections is passed to the light-curve analysis described in Section 5.

Figure 1 shows representative image cutouts of recovered asteroids. These examples illustrate the type of source detections included in the statistical analysis; all quantitative results below are derived from the full recovered sample.

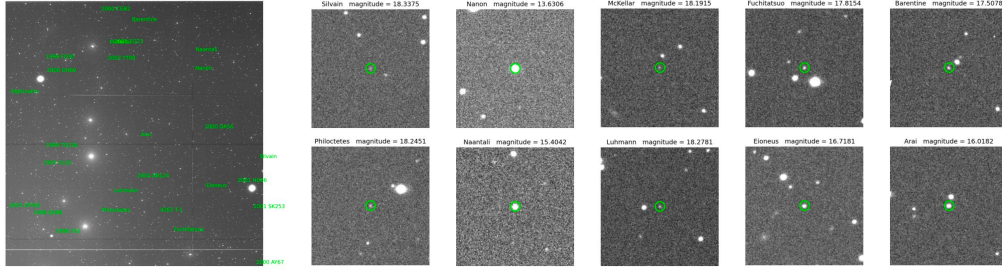


Fig. 1: Representative known-asteroid detections and image cutouts from the pilot-survey material. Green labels and circles identify recovered known asteroids. The figure illustrates the type of image-level detections represented by the accepted associations analyzed in this work.

3 KNOWN-ASTEROID EXTRACTION METHOD

3.1 Known-Object Association Strategy

The goal of the pipeline is to identify detections of cataloged asteroids in calibrated GOTTA Prototype source catalogs. The procedure is designed as a known-object association workflow rather than a discovery-oriented linking pipeline. For each exposure, the pipeline reads the calibrated photometric catalog and the corresponding FITS header. The WCS solution is used to determine the sky footprint of the detector, while the exposure midpoint is computed from the start time and exposure duration. Known-asteroid candidates are predicted for this epoch and for the Xinglong observer location. Objects outside the padded field region, outside the CCD footprint, or fainter than the adopted ephemeris-magnitude limit are removed before cross-matching. The remaining predictions are associated with catalog detections using an angular nearest-neighbor criterion. Likely stationary-source contaminants are rejected with a Gaia DR3 cross-check before the recovered detections are merged with source photometry and augmented with orbit and observing-geometry quantities from external small-body databases.

The statistics presented in this paper are produced by applying the association workflow to the accumulated prototype data set in batch mode. In parallel, we have also implemented a single-

night processing version for future survey operations. The two modes share the same core steps—ephemeris prediction, footprint filtering, catalog association, Gaia-based stationary-source rejection, and small-body augmentation—but differ in their execution granularity. The batch mode is used here to obtain homogeneous statistics over the full data set, whereas the single-night mode is designed to produce nightly known-object products during routine operations. Figure 2 illustrates this single-night implementation. The present analysis uses only recovered detections and is therefore appropriate for evaluating the quality and content of the known-object sample, but not for measuring a formal detection efficiency, which would require the full set of predicted but unrecovered objects.

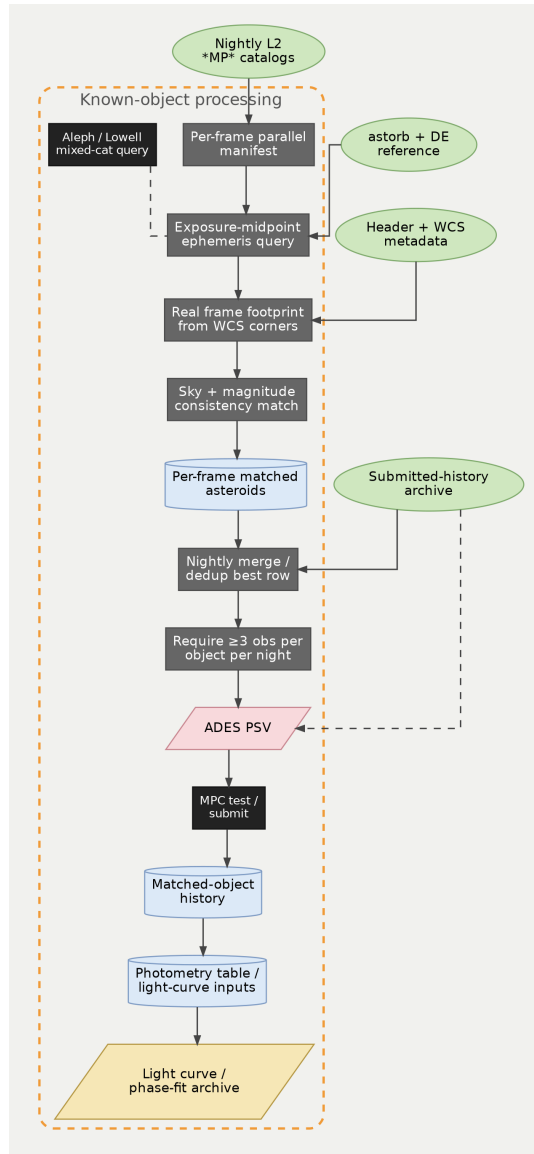


Fig. 2: Single-night workflow of the known-asteroid extraction procedure. The workflow starts from calibrated per-exposure photometric catalogs and exposure metadata, computes the exposure geometry and observing epoch, predicts known-object positions, applies field and footprint filters, matches surviving predictions to catalog detections, rejects stationary Gaia-source contaminants, and augments accepted associations with external small-body information.

3.2 Ephemeris Prediction and Exposure Geometry

For each cataloged asteroid, the ephemeris calculation starts from osculating orbital elements provided by a minor-planet orbital-element catalog, such as the Lowell Observatory astorb database or the standard MPC small-body orbit records. The elements specify the heliocentric orbit at a reference epoch. In the two-body approximation, the mean anomaly is advanced from the reference epoch to the exposure midpoint using

$$M(t) = M_0 + n(t - t_0), \quad (1)$$

where $n = \sqrt{\mu_\odot/a^3}$ is the mean motion. Kepler's equation,

$$M = E - e \sin E, \quad (2)$$

is then solved for the eccentric anomaly E . The position in the orbital plane is

$$x' = a(\cos E - e), \quad y' = a\sqrt{1 - e^2} \sin E. \quad (3)$$

This vector is rotated into the heliocentric ecliptic frame using the argument of perihelion, inclination, and longitude of the ascending node. The topocentric direction is obtained by subtracting the observer position at the same epoch and transforming the resulting vector into the equatorial frame. The apparent right ascension and declination follow from this topocentric line-of-sight vector.

This two-body description summarizes the physical basis of the screening calculation. In practice, high-precision apparent ephemerides also require time-scale handling, light-time correction, observer location, and perturbations from major Solar System bodies. The fast screening step therefore uses orbital elements from a minor-planet orbit catalog, while the final accepted detections are augmented with database and observer-dependent quantities from the JPL Small-Body Database¹ and Horizons² systems. The Minor Planet Center³ provides the broader astrometric-observation infrastructure and reporting context for Solar System small-body work. The Lowell Observatory astorb database⁴ is used as an orbit-catalog resource for fast screening.

The exposure footprint is computed directly from the WCS of each calibrated catalog. The detector corners are projected from pixel coordinates to the sky, and the search radius is initialized from the maximum angular distance between the field center and the projected corners, with an additional padding term to avoid edge losses. The exposure midpoint is

$$t_{\text{mid}} = t_{\text{start}} + \frac{1}{2}t_{\text{exp}}. \quad (4)$$

This midpoint is used for ephemeris evaluation because even modest sky-plane motion can lead to measurable offsets if the start time is used instead. After the initial sky-coordinate query, predicted positions are transformed back to detector coordinates, and only candidates falling within the valid pixel range are retained.

¹ https://ssd.jpl.nasa.gov/tools/sbdb_lookup.html

² <https://ssd.jpl.nasa.gov/horizons/>

³ <https://minorplanetcenter.net/>

⁴ <https://asteroid.lowell.edu/main/astorb/>

3.3 Catalog Matching and Small-Body Augmentation

The filtered predictions are then associated with the GOTTA source catalog. For each predicted position, the nearest catalog source is identified within the adopted arcsecond-level matching radius. This radius is chosen to be larger than the expected centroiding uncertainty of faint moving sources but small enough to suppress random associations in crowded fields. The observed-minus-predicted residuals reported below are computed from the small-angle offsets in right ascension and declination after the accepted matches have passed the Gaia stationary-source filter. The angular separation is computed on the sphere, with the right-ascension difference projected by $\cos \delta$ for small-angle residuals.

A Gaia-based stationary-source mask is applied after the asteroid-source association. A predicted asteroid position may occasionally fall close to a field star, in which case the nearest-neighbor source in the GOTTA catalog can be the stationary star rather than the moving object. To suppress this contamination, every accepted asteroid-source association is cross-matched with Gaia DR3 (Gaia Collaboration et al. 2023). If the same GOTTA source is also matched to a Gaia source within 1.5 arcsec, the association is rejected from the asteroid sample. This simple mask removes obvious stellar contaminants before the astrometric residuals are analyzed as a function of brightness and sky-plane angular rate. This step is applied before all residual and rate-dependent statistics reported below.

For each accepted association, the measured catalog quantities are merged with the predicted ephemeris. The photometric measurements include aperture magnitudes and their uncertainties; Kron-like measurements are retained for comparison but are not used as the primary photometric statistic in the final analysis. The predicted asteroid magnitude is used only as a consistency diagnostic, because it is not necessarily defined in the same photometric system as the GOTTA *g*-band measurement and is also affected by asteroid color, phase behavior, and rotational variability.

The matched detections are further associated with small-body database entries to obtain object identifiers, orbit classes, orbital elements, and observer-dependent geometry. The latter includes heliocentric distance, topocentric distance, phase angle, and sky-plane angular rates. These quantities are used in the statistical analysis below to interpret the recovered sample in terms of orbital population, observing geometry, and apparent motion.

4 RESULTS

4.1 Recovered Sample, Orbital Populations, and Observing Geometry

The recovery procedure yields 56,230 accepted associations corresponding to 16,053 distinct known asteroids. These detections span 73 UTC nights during the 2025 January–June prototype campaign. The size of this sample is sufficient to characterize the dominant orbital populations recovered by the survey and to examine how the association residuals depend on brightness and apparent motion. Table 1 summarizes the main statistics.

The orbit-class composition is shown in Table 2. Main-belt asteroids account for 51,058 detections and 14,674 unique objects. Outer main-belt asteroids, Jupiter Trojans, inner main-belt

Table 1: Statistics for the recovered GOTTA known-asteroid sample.

Quantity	Value	Note
Accepted source-ephemeris associations	56,230	Accepted source-ephemeris associations
Distinct known asteroids	16,053	Unique minor-planet identifiers
Exposure catalogs with accepted associations	13,253	Per-exposure catalog products
UTC nights represented	73	Recovered sample date coverage
Field groups represented	588	Grouped by pointing metadata
NEO-class objects	77	Identified from small-body metadata
PHA-class objects	19	Identified from small-body metadata
Median ephemeris magnitude	18.2215	Predicted brightness
Median g_{aper}	18.1110	Adopted aperture magnitude
Median $\sigma(g_{\text{aper}})$	0.1239 mag	Adopted aperture uncertainty
Median aperture S/N proxy	8.7621	Flux/error in adopted aperture
Median observed-minus-predicted separation	0.220 arcsec	Recovered detections
84th-percentile observed-minus-predicted separation	0.484 arcsec	Recovered detections
2D residual RMS	0.355 arcsec	RMS of the 2D separation
RMS in $\Delta\alpha \cos \delta$	0.250 arcsec	Coordinate residual
RMS in $\Delta\delta$	0.252 arcsec	Coordinate residual
Median angular rate	29.276 arcsec hr ⁻¹	Derived from augmented geometry
Median phase angle	13.245 deg	Derived from augmented geometry
Associations without geometry augmentation	107	Excluded from geometry statistics

asteroids, and Mars-crossing asteroids make up most of the remaining sample. Classes with only a few detections are grouped as Other/unclassified, including the generic SBDB class AST, which denotes an asteroid not assigned to a more specific dynamical class in the adopted mapping. The NEO classes are represented by Amor, Apollo, and Aten objects. The dominance of main-belt objects is expected because the survey footprint and limiting depth preferentially sample objects near ordinary main-belt opposition geometries. The smaller NEO and PHA subsets should therefore be viewed as validation samples for the recovery path rather than as indicators of discovery performance.

The recovery rate also varies substantially from night to night because of sky coverage, observing conditions, and the number of available exposure catalogs. The five most productive UTC nights are listed in Table 3.

The recovered detections are concentrated in the surveyed sky regions available during the prototype campaign, reflecting the combination of the survey footprint, seasonal visibility, and the instantaneous distribution of known minor planets. The orbital distribution confirms that the sample is dominated by main-belt objects, with smaller contributions from inner and outer

Table 2: Orbit-class composition of the recovered known-asteroid sample. Separations are in arcsec. Classes with only a few detections are grouped into “Other/unclassified”. The original generic SBDB class “AST” denotes an asteroid without a more specific dynamical class in the adopted mapping.

Code	Orbit class	Detections	Unique objects	Median g_{aper}	Median separation
MBA	Main-belt Asteroid	51,058	14,674	18.123	0.221
OMB	Outer Main-belt Asteroid	1,970	522	17.893	0.187
TJN	Jupiter Trojan	1,362	317	18.005	0.207
IMB	Inner Main-belt Asteroid	897	290	18.315	0.257
MCA	Mars-crossing Asteroid	656	166	17.847	0.228
AMO	Amor	139	31	17.143	0.227
APO	Apollo	101	38	17.749	0.253
TNO	TransNeptunian Object	20	3	17.588	0.243
ATE	Aten	19	8	17.668	0.292
Other	Other / unclassified	8	4	18.223	0.218

Table 3: Five UTC nights with the largest numbers of accepted known-asteroid associations.

UTC date	Detections	Unique objects	Exposure catalogs	Median g_{aper}	Median separation
2025-02-05	3,385	3,362	353	18.739	0.187
2025-01-19	3,119	3,065	576	18.354	0.189
2025-03-05	2,657	2,639	372	18.439	0.220
2025-01-28	2,656	2,638	395	18.499	0.189
2025-01-29	2,645	2,621	307	18.498	0.201

main-belt asteroids, Jupiter Trojans, Mars-crossers, and NEOs. These distributions are shown in Figures 3 and 4.

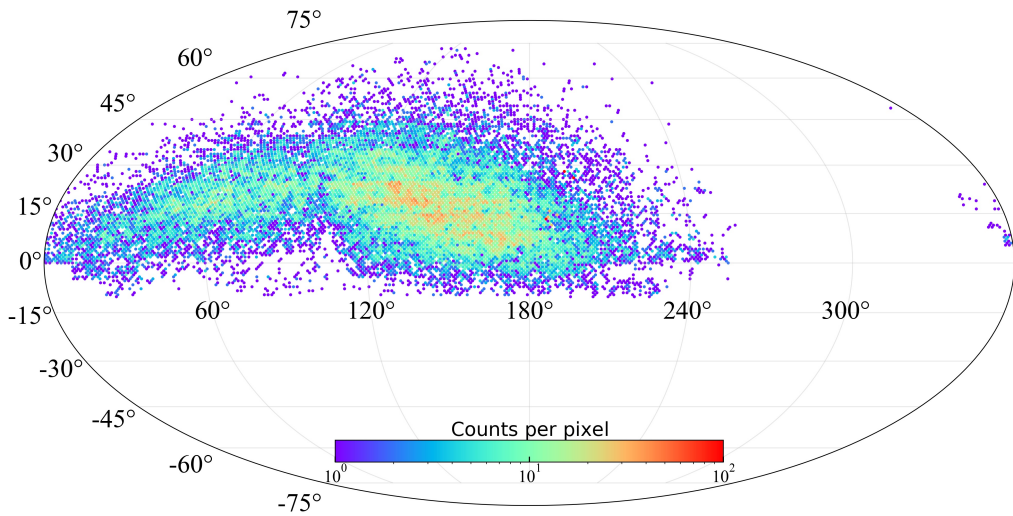


Fig. 3: Detection-level sky distribution of recovered known asteroids in equatorial coordinates, binned with HEALPix nside 64. The color scale is clipped at 100 detections per pixel for readability.

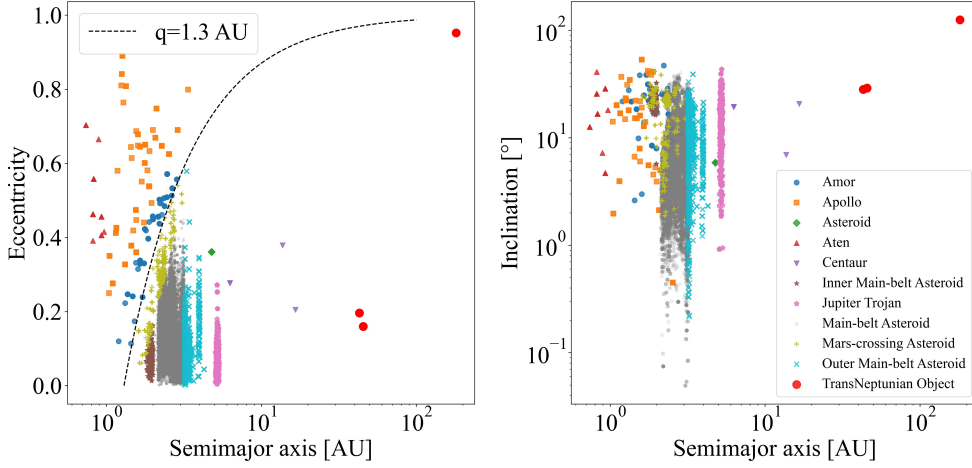


Fig. 4: Orbital-element distribution of the recovered known-asteroid sample. Points are colored by orbit class. The dashed curve in the semi-major-axis–eccentricity panel marks perihelion distance $q = 1.3$ AU, a conventional boundary for near-Earth objects.

The corresponding observing geometry is summarized in Figure 5. The median heliocentric distance is 2.581 AU, the median topocentric distance is 1.840 AU, and the median phase angle is 13.245 deg. These values are consistent with a sample dominated by ordinary main-belt observing geometries and explain why the main angular-rate distribution is relatively narrow. Although the sample is dominated by main-belt asteroids, the recovered NEO/PHA subset provides a useful stress test because these objects tend to have higher apparent rates and more variable observing geometry. The NEO subset contains 259 detections of 77 unique objects, with median $g_{\text{aper}} = 17.525$ mag, median angular rate $111.643 \text{ arcsec hr}^{-1}$, and median residual separation 0.232 arcsec . The PHA subset contains 53 detections of 19 unique objects, with median $g_{\text{aper}} = 17.660$ mag, median angular rate $132.424 \text{ arcsec hr}^{-1}$, and median residual separation 0.211 arcsec . These small subsets are not large enough for a formal population-level performance assessment, but they show that the current association path already covers higher-rate Solar System targets.

Having established that the recovered catalog is dominated by main-belt objects observed at ordinary main-belt geometries, we next examine whether the measured brightness and photometric uncertainties are consistent with a sample of mostly faint single-exposure asteroid detections.

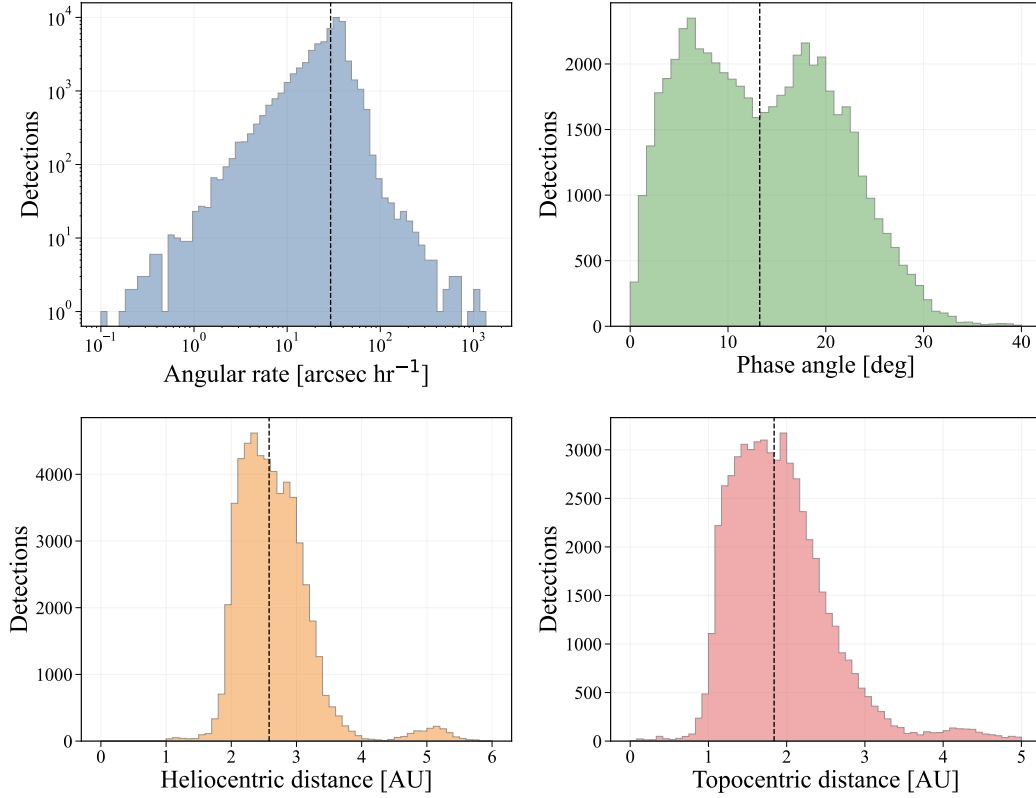


Fig. 5: Observing-geometry diagnostics for recovered known asteroids with successful geometry augmentation. The panels show sky-plane angular rate, phase angle, heliocentric distance, and topocentric distance. Vertical dashed lines mark the corresponding medians.

4.2 Photometric Properties of the Recovered Asteroids

For the photometric analysis we use the aperture magnitude corresponding to the optimal aperture defined by the basic GOTTA photometric pipeline. In that pipeline, aperture photometry is measured in a series of circular apertures, and the optimal aperture is determined from the signal-to-noise ratio of stellar sources as a function of aperture radius. A larger reference aperture is used for aperture correction, so that the cataloged aperture magnitudes are placed on a consistent photometric scale. We therefore do not re-derive an asteroid-specific optimal aperture in this work. Instead, we use the pipeline-defined optimal aperture as a homogeneous measurement for the recovered asteroid sample.

This choice is preferable to Kron-like photometry for the statistical analysis presented here. Kron apertures are adaptive and can respond differently to background noise, faint-source segmentation, and possible source elongation, whereas the adopted aperture magnitude provides a uniform measurement across the sample. Kron photometry is retained in the catalog for comparison, but it is not used as the primary quantity in the magnitude, uncertainty, and astrometric-residual diagnostics below.

We denote the adopted aperture magnitude as g_{aper} below. Most recovered asteroids lie close to the faint end of routine single-exposure catalog measurements rather than in the high signal-to-noise regime used for stellar calibration. The uncertainty–magnitude relation increases toward

the faint end, as expected from decreasing source signal-to-noise ratio. The difference between predicted ephemeris magnitude and measured GOTTA g -band aperture magnitude is used only as a consistency diagnostic, because the two quantities are not necessarily defined in the same photometric system and may also differ because of asteroid color, phase behavior, rotation, shape, and ephemeris-magnitude uncertainties. The corresponding photometric distributions are summarized in Figure 6.

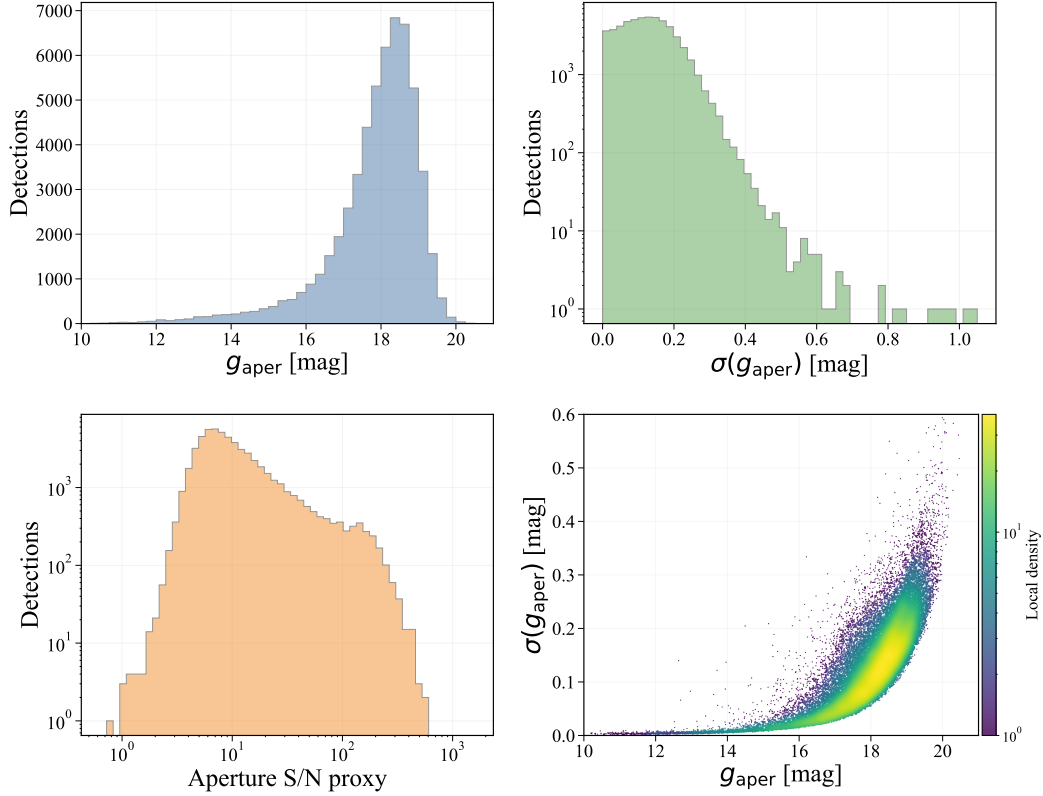


Fig. 6: Aperture-photometry diagnostics for recovered known asteroids. The adopted aperture follows the optimal aperture defined by the GOTTA basic photometric pipeline. The panels show the adopted aperture magnitude distribution, the corresponding magnitude-uncertainty distribution, the aperture S/N proxy, and the dependence of magnitude uncertainty on adopted aperture magnitude. The last panel is shown as a density map to emphasize the dominant distribution of faint asteroid detections.

Because the photometric uncertainty rises rapidly toward the faint end, the same magnitude dependence is expected to appear in the astrometric residuals through centroiding uncertainty.

4.3 Astrometric Residuals and Apparent Motion

The residual distribution is strongly concentrated below the adopted matching scale, with a median separation of 0.220 arcsec and an 84th percentile of 0.484 arcsec. The two-dimensional RMS is 0.355 arcsec, and the RMS values of $\Delta\alpha \cos \delta$ and $\Delta\delta$ are 0.250 arcsec and 0.252 arcsec, respectively. The coordinate residuals are centered close to zero, indicating that the accepted associations do not show a large global offset between the predicted ephemerides and the calibrated source positions.

When the residuals are grouped by adopted aperture magnitude, the median separation increases from about 0.09–0.11 arcsec for relatively bright detections to about 0.27 arcsec near $g_{\text{aper}} \simeq 18$ –19 mag. This trend indicates that source centroiding and signal-to-noise ratio dominate the residual budget for most recovered asteroids.

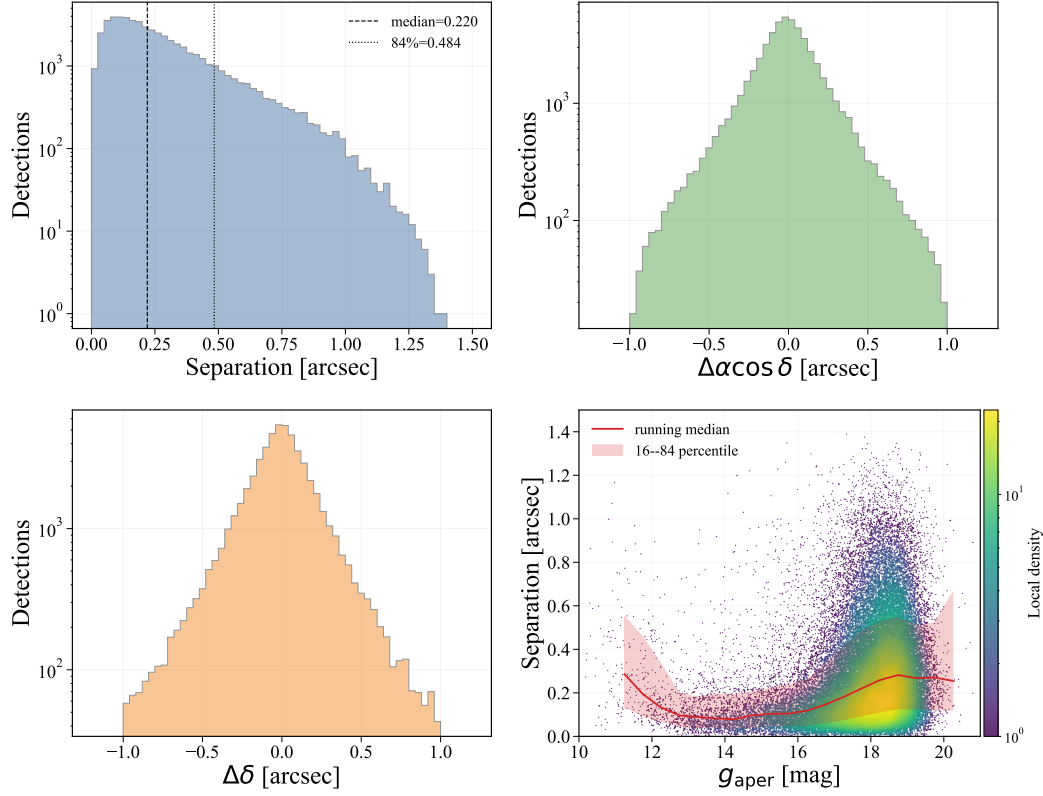


Fig. 7: Astrometric residual diagnostics. The panels show the observed-minus-predicted separation, $\Delta\alpha \cos \delta$, $\Delta\delta$, and separation as a function of the adopted aperture magnitude g_{aper} . In the lower-right panel, the solid curve and shaded region indicate the running median and 16th–84th percentile range in magnitude bins.

The dependence on sky-plane angular rate is weaker over the main part of the sample. Most recovered objects have moderate rates, with a median of $29.276 \text{ arcsec hr}^{-1}$. In this regime, the median separation remains close to the full-sample value. The sparsely populated high-rate tail shows larger scatter and should be interpreted with caution because it contains few detections and may be more affected by trailing, centroiding biases, or matching ambiguity.

The magnitude and rate dependences are shown in Figures 7 and 8, where the running medians summarize the main trends without introducing additional binned tables.

These residuals should be interpreted as association-level astrometric diagnostics rather than as a direct measurement of orbit improvement. For most well-observed main-belt asteroids, the predicted ephemeris uncertainty is expected to be smaller than the centroiding uncertainty of the faint GOTTA detections, so the observed-minus-predicted residual mainly traces the positional precision of the recovered survey source. However, no orbit refitting is performed in this work. The results therefore demonstrate that the prototype catalogs can provide sub-arcsecond asteroid

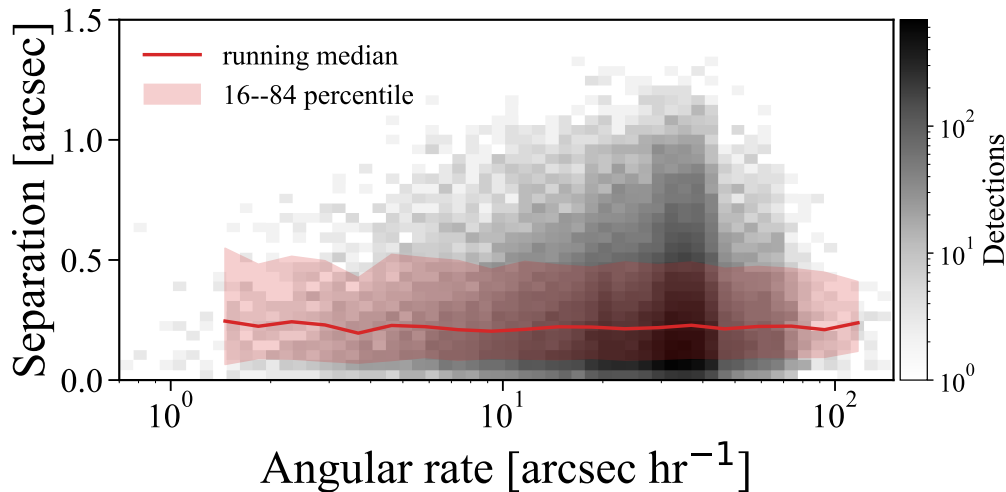


Fig. 8: Observed-minus-predicted separation as a function of sky-plane angular rate. The gray density map shows the dominant range of the recovered sample, while the red curve and shaded region show the running median and the 16th–84th percentile range in logarithmic rate bins. Most detections lie at moderate angular rates, where the residual distribution remains close to the full-sample behavior. A small number of higher-rate detections are excluded from the displayed range for clarity but are included in the binned statistics.

astrometry suitable for known-object association and potential future orbit-refinement experiments, but they do not by themselves quantify the improvement of minor-planet orbits.

The weak residual dependence on angular rate indicates that, for most detections, the sample is limited more by brightness and centroiding precision than by apparent motion. We therefore turn next to the observing geometry and revisit statistics that determine the time-domain usefulness of the recovered objects.

4.4 Temporal Sampling and Light-Curve Potential

The scientific value of a high-cadence survey for asteroid work depends not only on the number of recovered objects, but also on how often individual objects are revisited. In the current sample the revisit statistics are highly non-uniform. The median number of detections per object is two, the mean is 3.50, and the 84th percentile is six. A small subset of objects has much denser sampling; the most frequently recovered object has 317 detections over nine distinct nights. These repeatedly recovered objects are useful for time-series construction tests, although they are not representative of the full sample. Table 4 lists the five most frequently recovered objects.

These repeated detections are useful for constructing sparse asteroid time series and for checking the internal consistency of the photometry. However, the prototype-only sampling is not sufficient for robust period determination for most objects, as shown by the cumulative detection-count distribution in Figure 9. This motivates the auxiliary light-curve experiment in Section 5, where the GOTTA detections provide part of the input time series and the nearby 60 cm Schmidt telescope supplies denser temporal coverage for selected targets.

Table 4: Five most frequently recovered known asteroids in the GOTTA Prototype sample.

Object ID	Name	Detections	Nights	First MJD	Last MJD	Baseline (days)	Median g_{aper}	Class
214386	2005 MU1	317	9	60739.9	60818.7	78.824	18.141	MBA
76719	2000 JJ18	256	3	60738.8	60798.8	59.974	18.595	MBA
293	Brasilia	67	14	60709.9	60809.6	99.701	15.233	MBA
15751	1991 VN4	47	9	60714.8	60809.6	94.840	17.684	MBA
33629	1999 JK76	40	13	60709.9	60818.7	108.732	17.859	MBA

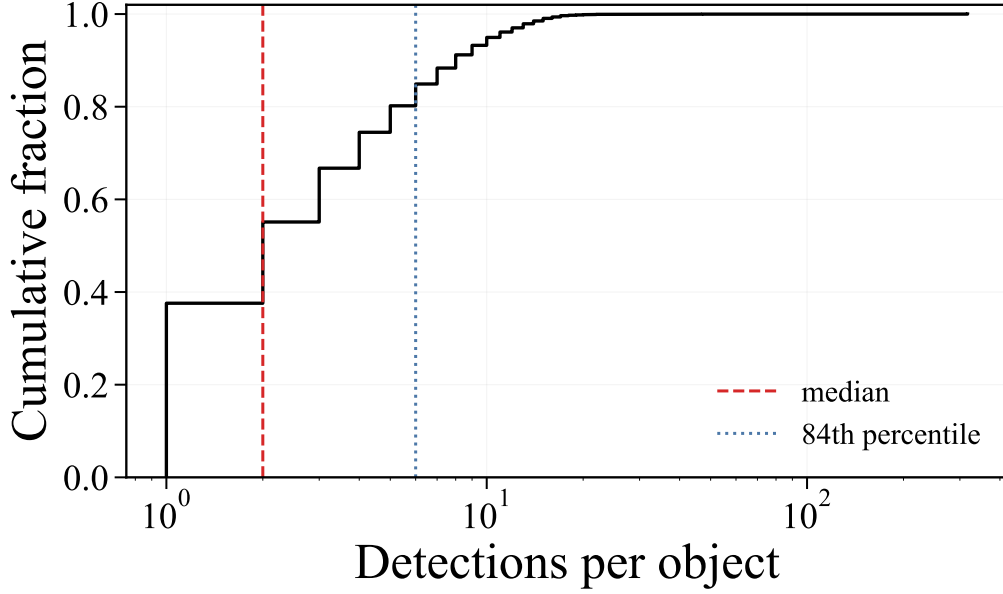


Fig. 9: Cumulative distribution of the number of recovered detections per object. Most objects are recovered only a few times, while a small tail has tens to hundreds of detections and is useful for light-curve pipeline tests.

5 LIGHT-CURVE ANALYSIS WITH AUXILIARY OBSERVATIONS

5.1 Motivation and Input Data

Asteroid light curves provide rotation periods, amplitudes, and basic constraints on shape and spin state, and they are therefore an essential downstream product of a moving-object survey pipeline. The known-object recovery described above supplies object identities, observing epochs, and calibrated photometry, but the prototype survey cadence alone does not provide sufficient phase coverage for most targets. We therefore use auxiliary observations from a nearby 60 cm Schmidt telescope to test the period-search workflow on selected asteroids. The combined data set is used as a validation sample rather than as a complete rotational survey. GOTTA Prototype measurements provide part of the time series and the object-association framework, while the 60 cm Schmidt observations provide denser sampling for the selected targets.

For each target, we construct light curves from the available photometric measurements and apply quality control separately to aperture, PSF, and Kron-like photometry. Measurements with

missing values or obvious outliers are removed, and a light curve is retained for period analysis only if it contains at least 60 valid points. This requirement reduces the number of spurious periodogram peaks caused by sparse sampling and provides a minimum phase coverage for model comparison. When multiple photometric measurements are available for the same target, they are treated as independent checks on the candidate period. Only data sets with negligible inter-night or inter-instrument zero-point offsets are used in the current demonstration. For the short time spans considered here, changes in heliocentric distance and topocentric distance are small for most selected targets; the pipeline nevertheless records the observing geometry so that reduced magnitudes or phase corrections can be applied in future analyses.

5.2 Period Search and Model Selection

Candidate periods are searched using three complementary methods. The Lomb–Scargle periodogram is used as the primary method for unevenly sampled time-series data (Lomb 1976; Scargle 1982). Phase Dispersion Minimization (PDM) provides an independent search that is less sensitive to the assumption of a sinusoidal light-curve shape and is therefore useful for asteroid light curves (Stellingwerf 1978; Warner et al. 2009). A frequency-domain diagnostic based on a Fourier transform of an interpolated, uniformly sampled light curve is also used as an auxiliary check. The initial search focuses on candidate periods shorter than 10 hr; adopted double-peaked rotation periods may therefore extend to approximately 20 hr.

Candidate periods from the different search methods and photometric measurements are grouped in relative-period space. Periods that agree within 10 per cent are assigned to the same cluster, and the median period of the largest cluster is used as the initial candidate period. If no repeated period cluster is found, the object is not assigned a robust period at this stage. This method avoids relying on a single periodogram peak and provides a simple consistency requirement across photometric measurements and search algorithms. We record this agreement through a match-point metric across the Lomb–Scargle, PDM, and FFT searches; the intermediate search-quality flag in the period-analysis sample is based on this metric and records the consistency of the period-search candidates before the final folded-light-curve assessment. The initial clustering is performed without using an external reference period; literature periods are consulted only as a later comparison.

Asteroid light curves often show double-peaked morphology because the projected shape generally repeats twice during one rotation. We therefore test both P and $2P$ for each candidate period. For double-peaked light curves, the adopted rotation period is the double-peaked period. The phase is computed as

$$\phi = \left[\frac{t - t_0}{P} \right] \bmod 1, \quad (5)$$

where t_0 is an arbitrary reference epoch. The folded light curve is fitted with a Fourier series,

$$m(\phi) = a_0 + \sum_{k=1}^N [a_k \cos(2\pi k\phi) + b_k \sin(2\pi k\phi)], \quad (6)$$

where $N \leq 3$ in the current implementation. The fit is performed with photometric-error weighting and iterative sigma clipping to reduce the influence of outliers. The model order and the choice between P and $2P$ are evaluated using the residuals and the Bayesian Information Criterion (Schwarz 1978),

$$\text{BIC} = n \ln(\sigma^2) + k \ln n, \quad (7)$$

where n is the number of valid measurements, k is the number of model parameters, and σ is the residual scatter.

To identify plausible double-peaked solutions, we also examine the harmonic content and morphology of the Fourier model. In particular, the relative strength of the second harmonic,

$$R_{21} = \frac{A_2}{A_1}, \quad (8)$$

is used together with the number of peaks, phase coverage, phase-overlap statistics, and the residual improvement between P and $2P$. High-confidence solutions are visually inspected to reject obvious aliases and poorly covered folded light curves. For accepted solutions, the period uncertainty is estimated from a local scan of the fit statistic around the adopted period.

5.3 Candidate Classification and Preliminary Results

The combined GOTTA and 60 cm Schmidt data are used to identify asteroid light curves with plausible rotational modulation. The updated analysis retains 24 objects in the period-analysis validation sample. Twelve objects are assigned reliable final quality flags and are summarized in Table 5; the remaining twelve are retained as tentative candidates requiring additional observations. Within the external period database queried in this work, none of the retained candidates has a matched reference period. We therefore treat these periods as validation-sample measurements without claiming a complete census of previously unknown rotation periods.

Appendix Figures A.1 and A.2 show folded light curves for the 24 objects retained in the updated validation sample, separated by final quality flag so that each mosaic can be displayed at full-page scale. The panel titles list the initial period and the adopted single- or double-peaked period used for the displayed phase folding. The black curves show the Fourier models used for the morphology check. The full period-analysis sample, including tentative solutions retained for follow-up, is given in Appendix Table A.1. These results demonstrate that, although the GOTTA Prototype data alone are generally too sparse for period determination, the recovered known-object catalog can be combined with auxiliary observations to validate the asteroid period-analysis pipeline.

Table 5: Reliable asteroid rotation-period measurements from the combined GOTTA Prototype and 60 cm Schmidt validation sample. Only objects assigned reliable final quality flags are included. The full validation sample, including tentative solutions, is given in Appendix A.

Object ID	Type	N_{eff}	Span	P_{rot}	ΔP	Model	Phot.
2175	MBA	671	123.4	4.8880	0.0010	P	aper.
4479	MBA	1454	195.9	3.0055	0.0197	P	aper.
5424	MBA	722	99.9	5.4887	0.0014	$2P$	Kron
7531	MBA	503	49.0	3.3375	0.0004	$2P$	PSF
12148	MBA	457	29.5	4.5248	0.0104	$2P$	aper.
19376	MBA	1066	196.5	5.4896	0.0001	P	aper.
22542	MBA	190	23.9	2.8439	0.0070	P	Kron
25745	MBA	1540	170.8	5.5143	< 0.0001	$2P$	PSF
46082	MBA	1181	171.6	5.6907	0.0013	P	PSF
72918	MBA	1466	221.9	6.1385	0.0004	P	PSF
79589	MBA	443	50.2	3.3634	0.0098	P	aper.
97776	MBA	587	78.1	3.5371	0.0417	$2P$	PSF

6 DISCUSSION AND CONCLUSIONS

The recovered known-asteroid sample demonstrates that calibrated GOTTA Prototype catalogs can support systematic association with cataloged Solar System objects. The main validation provided by this work is the closure of the known-object association workflow: calibrated source catalogs are combined with exposure geometry and mid-exposure timing, known-object ephemerides are filtered by the real CCD footprint, likely stationary-source contaminants are removed using Gaia cross-checks, accepted detections are matched in the source catalog, and the resulting associations are augmented with small-body database information. This workflow is a prerequisite for future moving-object products, including candidate-level bookkeeping, tracklet construction, and moving-object-aware transient filtering.

Known-object recovery is a useful validation mode because the predicted positions of cataloged asteroids are obtained independently of the GOTTA source extraction. The observed-minus-predicted residuals therefore test the combined effects of WCS quality, timing accuracy, centroiding, ephemeris uncertainty, and matching behavior. The median separation of 0.220 arcsec and the two-dimensional RMS of 0.355 arcsec show that the accepted associations are well localized on the scale required for known-object recovery. The increase of residual with fainter aperture magnitude indicates that centroiding precision and signal-to-noise ratio dominate the residual distribution for most of the sample. The weak dependence on angular rate over the main rate range suggests that trailing is not the primary limitation for the bulk of recovered objects, although the sparsely populated high-rate tail still requires dedicated analysis.

The residual analysis also clarifies what can and cannot be inferred about orbit refinement. For the majority of well-known main-belt objects, the ephemeris uncertainty is expected to be

small compared with the centroiding uncertainty of faint GOTTA detections, so the observed-minus-predicted residuals provide a useful proxy for the astrometric quality of recovered moving sources. However, this work does not refit asteroid orbits using the GOTTA measurements. The results therefore do not quantify orbit improvement directly. They instead show that the prototype can deliver sub-arcsecond asteroid positions that are suitable for future astrometric reporting and orbit-refinement experiments, particularly for NEOs, PHAs, and objects with shorter observational arcs.

The photometric results should be interpreted as survey-catalog diagnostics rather than asteroid physical measurements. The adopted aperture magnitude provides a homogeneous brightness estimate, but the ephemeris magnitude is not a calibrated reference in the GOTTA g band. Asteroid color, rotational modulation, phase-angle dependence, shape, and uncertainties in the magnitude model can all contribute to the difference between predicted and observed brightness. The light-curve analysis in Section 5 illustrates how the recovered known-object catalog can be used as the input layer for asteroid time-series work. The GOTTA-only sampling is generally too sparse for period determination, but the addition of 60 cm Schmidt observations allows the period-search machinery to be tested on selected targets. The reliable periods obtained from the combined data should therefore be interpreted as a pipeline validation sample rather than as a complete rotational census.

Several limitations should be kept in mind when interpreting the present recovered sample. Because the catalog contains only accepted associations, it does not provide the denominator required for a formal recovery-efficiency estimate. The Gaia cross-check removes obvious stationary-source contaminants, but it does not by itself provide a complete false-association model. A quantitative false-match rate still requires the full set of predicted candidates, rejected associations, and original per-exposure source catalogs, together with random-shift or time-scrambling tests. A quantitative trailing analysis will also require exposure-level seeing and exposure-time information, so that the expected trail length,

$$L_{\text{trail}} = \mu t_{\text{exp}}, \quad (9)$$

can be compared with astrometric residuals and photometric uncertainties. Here μ is the sky-plane angular rate and t_{exp} is the exposure time.

For the future GOTTA array, the strength of the system will lie in the combination of large sky coverage and frequent temporal sampling. The prototype results show that the association and augmentation framework needed for known-object recovery can be implemented on GOTTA data. For the full GOTTA array, denser revisit patterns and multi-site coverage should increase the number of objects with sufficient phase coverage for period recovery, reducing the need for auxiliary observations for many targets. With candidate-level bookkeeping, repeated field visits, multi-band observations, and moving-object-aware scheduling, the same framework can be extended toward NEO monitoring, asteroid light-curve studies, and eventually discovery-oriented moving-object processing.

In summary, we have developed and tested a known-asteroid recovery workflow for GOTTA Prototype photometric catalogs. The recovered sample contains 56,230 detections of 16,053 unique known asteroids over 73 UTC nights, including small subsets of NEOs and PHAs. The sample is dominated by main-belt asteroids and provides statistically useful distributions in sky position, orbital class, aperture magnitude, angular rate, observing geometry, astrometric residual, and repeated temporal sampling. The sub-arcsecond observed-minus-predicted residuals demonstrate that the current catalog products are suitable for known-object association. The pilot light-curve analysis identifies 12 reliable rotation-period measurements from the combined GOTTA and 60 cm Schmidt validation sample, together with 12 tentative candidates requiring follow-up. This work establishes the first moving-object validation layer for the GOTTA Prototype and provides the basis for future candidate-level detection, tracklet construction, and asteroid light-curve analysis.

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Software: Astropy ([Astropy Collaboration et al. 2013, 2018, 2022](#)), NumPy ([Harris et al. 2020](#)), SciPy ([Virtanen et al. 2020](#)), scikit-learn ([Pedregosa et al. 2011](#)), Matplotlib ([Hunter 2007](#)), pandas ([McKinney 2010](#)), astroquery ([Ginsburg et al. 2019](#)), and healpy/HEALPix ([Górski et al. 2005](#)).

Appendix A: LIGHT-CURVE VALIDATION PRODUCTS

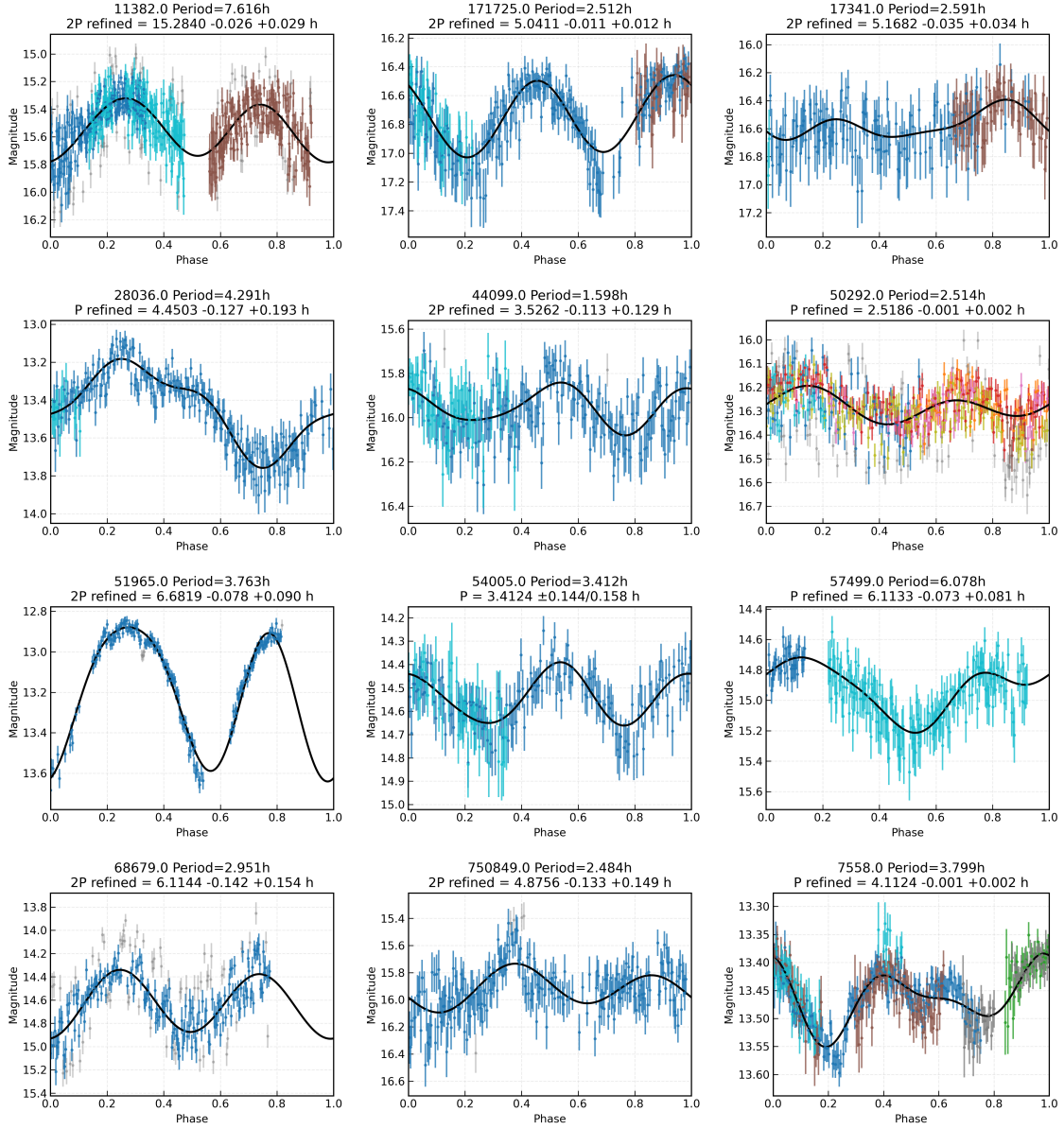


Fig. A.1: Phase-folded light curves for the 12 objects assigned reliable final quality flags in the updated combined GOTTA Prototype and 60 cm Schmidt validation sample. Colored points denote retained measurements from different data subsets, gray points mark rejected or lower-weighted measurements, and black curves show the adopted Fourier models. Each panel gives the object identifier and the candidate period information used in the current period analysis.

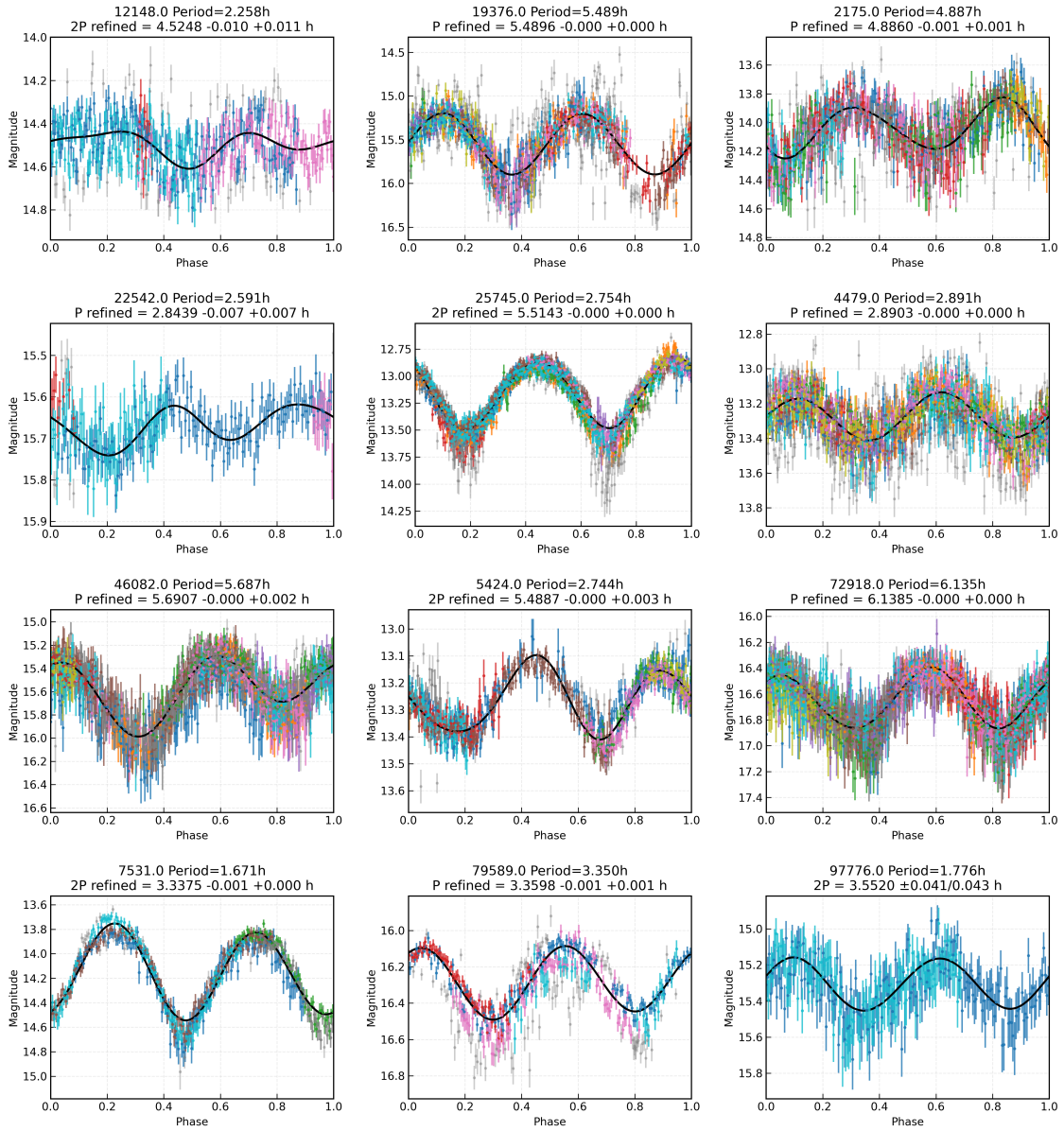


Fig. A.2: Same as Figure A.1, but for the 12 objects assigned tentative final quality flags. These solutions are retained in the validation sample but require additional observations for confirmation.

Table A.1: Final period-analysis table for the combined GOTTA Prototype and 60 cm Schmidt validation sample. The column “Type” gives the small-body orbit class used in the known-asteroid catalog. N_{total} is the total point count reported in the updated period table, and N_{eff} is the effective number of photometric points retained by the period-analysis quality cuts. The search-quality flag records the agreement among the Lomb–Scargle, PDM, and FFT period searches, while the final-quality flag is assigned after the folded-light-curve sampling and period-span assessment. Tent. denotes tentative follow-up candidates.

Object ID	H	Type	N_{total}	N_{eff}	Span	P_{rot}	ΔP	Model	Phot.	MP	Search	Final
2175	14.42	MBA	796	671	123.4	4.8880	0.0010	P	aper.	4	Rel.	Rel.
4479	13.26	MBA	1462	1454	195.9	3.0055	0.0197	P	aper.	7	Rel.	Rel.
5424	13.57	MBA	757	722	99.9	5.4887	0.0014	$2P$	Kron	3	Good	Rel.
7531	14.07	MBA	503	503	49.0	3.3375	0.0004	$2P$	PSF	6	Rel.	Rel.
12148	14.50	MBA	459	457	29.5	4.5248	0.0104	$2P$	aper.	5	Rel.	Rel.
19376	15.18	MBA	1093	1066	196.5	5.4896	0.0001	P	aper.	6	Rel.	Rel.
22542	15.97	MBA	229	190	23.9	2.8439	0.0070	P	Kron	2	Poss.	Rel.
25745	13.15	MBA	1541	1540	170.8	5.5143	< 0.0001	$2P$	PSF	6	Rel.	Rel.
46082	15.41	MBA	1211	1181	171.6	5.6907	0.0013	P	PSF	6	Rel.	Rel.
72918	16.66	MBA	1473	1466	221.9	6.1385	0.0004	P	PSF	6	Rel.	Rel.
79589	16.25	MBA	443	443	50.2	3.3634	0.0098	P	aper.	9	Rel.	Rel.
97776	15.31	MBA	588	587	78.1	3.5371	0.0417	$2P$	PSF	6	Rel.	Rel.
7558	13.45	MBA	438	392	51.3	4.1124	0.0019	P	PSF	–	Rel.	Tent.
11382	15.46	MBA	615	591	53.1	15.2840	0.0273	$2P$	aper.	6	Rel.	Tent.
17341	16.58	MBA	191	189	25.9	5.1682	0.0340	$2P$	PSF	4	Rel.	Tent.
28036	13.42	MBA	226	222	4.9	4.4503	0.1598	P	PSF	4	Rel.	Tent.
44099	16.02	MBA	216	209	4.8	3.5262	0.1210	$2P$	PSF	6	Rel.	Tent.
50292	16.29	MBA	522	496	53.4	2.5186	0.0015	P	aper.	4	Rel.	Tent.
51965	13.00	MBA	243	207	5.5	6.6819	0.0837	P	PSF	6	Rel.	Tent.
54005	14.54	MBA	810	760	124.6	3.5362	0.0936	P	PSF	4	Rel.	Tent.
57499	14.83	MBA	206	204	24.0	6.1133	0.0771	P	PSF	4	Rel.	Tent.
68679	14.56	MBA	244	243	4.7	6.1144	0.1479	$2P$	aper.	4	Rel.	Tent.
171725	16.69	MBA	286	276	26.3	5.0411	0.0118	$2P$	PSF	5	Rel.	Tent.
750849	15.94	MBA	215	210	4.8	4.8756	0.1409	$2P$	Kron	8	Rel.	Tent.

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